MIL-D-4827D
17 November 1972
Superseding
MIL-D-4827C
2 January 1967

MILITARY SPECIFICATION

DEVELOPER, PHOTOGRAPHIC, PAPER, ARMED FORCES DEVELOPER NO. 25

This specification is mandatory for use by all Departments and Agencies of the Departmentof Defense.

1. SCOPE

1.1 This specification covers the requirements of a processing formula for a photographic developer which is supplied in single powder dry form, properly stabilized, for use in the preparation of solutions primarily used in developing photographic papers.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-4672	Paper, Photographic, Black and White, Sensitized

Federal

O-C-275	Chemicals, Photographic, Single Substance, General Specifications for
PPP-P-16	Packaging of Photographic Dry Chemicals, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6750

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3. REQUIREMENTS

3.1 Chemicals. The chemicals used in the processing formula shall be photographic grade conforming to O-C-275. Materials used to prevent deterioration of the dry chemicals in the single-blend powder are not specifically designated but shall be entirely suitable for the purpose intended. These materials used shall be freely soluble in water, relatively nontoxic and photographically inert.

3.1.1 Mixing of chemicals. The dry chemicals in powder form shall be mixed thoroughly to form a homogeneous blend before dispensing into the unit container. The chemicals shall be free flowing, readily soluble, free from foreign matter, and shall show no signs of discoloration or dark specks. The temperature and the relative humidity of the mixing room shall not exceed 75° Farenheit (F) and 25 percent relative humidity.

* 3.2 Composition of formula. The developer shall consist of paper developing chemicals having the composition specified in Table I. This specification covers a single-blend powder for 1- and 5-gallon sizes. Unless otherwise specified, the allowable tolerance of the total net chemical weight in each unit container and the weight of the individual chemicals shall be ±3 percent. The supplier shall certify that the chemical composition of the developer furnished conforms to the appropriate formulation specified in Table I.

TABLE I. Composition of formula

Chemicals	1-gallon size (grams)	5-gallon size (grams)
Sodium sulfite (Anyhydrous)	170.0	850.0
Sodium carbonate monohydrate	302.8	1,514.0
Potassium bromide	7.2	36.0
p-methyl-paraminophenol sulphate (metol)	11.7	58.5
Hydroquinone $(C_6H_4(OH)_2)$	45.4	227.0
1/Total chemical weight	1/ 537.1	1/ 2,685.5

1/ To allow this formula to be prepared in a single powder form, it may be necessary to add additional water-soluble components, otherwise photographically inert, to prevent interaction or decomposition of the active components while in packaged form.

3.3 Performance. The developer shall perform equal to the control solution when tested as specified in 4.3.2.

* 3.4 Unit sizes. The chemicals shall be packaged in unit sizes sufficient to prepare working solutions of one, five, or more gallons as specified by the procuring activity (see 6.2). For packages other than those specified herein, the weight of the chemical composition of the formula shall be

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extrapolated accordingly. Example: To determine the quantity of sodium sulfite needed for a 20 gallon size of developer, multiply by 20 the amount needed for one gallon size; i.e., 20 x 170.0 grams = 3400 grams.

3.5 Storage stability. The dry powder, in the unit package, shall show no evidence of deterioration after storage for a period of 48 hours at 130° ± 2° F.

3.6 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the manufacture and packaging of photographic materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, sampling shall be performed in accordance with the provisions set forth in MIL-STD-105. For purposes of sampling, lots shall consist of all material of the same type and item submitted for acceptance and delivery at one time.

4.3 Testing of the end item. The photographic processing formula shall be tested for applicable characteristics as indicated in Table II for each lot presented for acceptance. The lot size shall be expressed in units of primary containers of developer. The sample unit shall be three unit containers. The inspection levels and acceptable quality levels shall be as indicated in Table II.

* 4.3.1 Dissolving test. The powder obtained from each sample shall be added to 80 percent of the final volume of distilled water at 90°F while stirring with sufficient agitation to keep the chemicals suspended. Stirring shall continue until chemicals are dissolved. Sufficient water shall be added to make final volume of developer and stirring shall continue until a uniform solution is obtained. The mixing process should not exceed 10 minutes. Ten minutes after the mixing, the chemical solution shall be clear. When cooled to 68° F, the developer solution shall be colorless to light pink in color. Although a slight flocculence of the cooled liquid is permissable, foreign matter in any form, rust, dirt or undissolved chemicals shall be cause for rejection.

* 4.3.2 Performance test. Prepare two solutions in distilled water, one of the sample and one of the control solution prepared according to the appropriate formula specified in Table I. Expose identical sensitometric images, containing at least eight steps, onto each of two strips of photographic paper taken from the same box or roll. Process one strip in each of the respective prepared solutions in accordance with the procedure described in MIL-P-4672. Compare the sensitometric characteristics of the paper

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TABLE II. Instructions for testing - sample units

Characteristic	Spec. reference		Rqmts. applied to		No. determinations per sample unit	Results reported as		Inspection level	AQL in defects per 100 units
	Requirement	Test method	Indiv. unit	Lot avg.		Pass or fail 1/	Numerically to nearest 2/		
Dissolving test	3.1	4.3.1	X	-	3	X	-	3/ S-1	4.0
Performance test	3.3	4.3.2							
Difference in paper speed			X	-	Avg. of 6		0.10	3/ S-1	4.0
Difference in partial scale			X	-	Avg. of 6	-	0.01	3/ S-1	4.0
Difference in base and fog density			X	-	Avg. of 6	-	0.01	3/ S-1	4.0
Difference in D max			X	-	Avg. of 6	-	0.01	3/ S-1	4.0
Storage stability	3.5	4.3.3							
Condition of powder			X	-	3	X	-	3/ S-1	4.0
Dust			X	-	3	X	-	3/ S-1	4.0
Performance									
Difference in paper speed			X	-	Avg. of 6	-	0.10	3/ S-1	4.0
Difference in partial scale			X	-	Avg. of 6	-	0.01	3/ S-1	4.0
Difference in base and fog density			X	-	Avg. of 6	-	0.01	3/ S-1	4.0
Difference in D max			X	-	Avg. of 6	-	0.01	3/ S-1	4.0

NOTE: 1/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.
2/ Test reports shall include all values on which results are based.
3/ Not less than 3 sample units shall be tested.

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developed in the solution prepared according to the specified formula. The partial scale of the curves prepared from paper developed in the test solution and the control solution shall not vary by more than ± .10, base and fog density shall not vary by more than ± .01, and D max shall not vary by more than ± .10. Tests shall be performed in duplicate on each unit. The average of the two determinations on each unit shall be used for plotting the sensitometric curve. Appropriate procedures for exposure of the paper and preparation of the sensitometric curve, as contained in MIL-P-4672 shall be used. Speed method selected for each identical processed sensitometric image shall not vary more than one-eighth of a stop by comparison.

* 4.3.3 Storage stability test. Unit containers selected at random from each lot shall be stored at 130°F for a period of 48 hours. The samples shall then be removed and allowed to cool to room temperature for 2 hours and then examined. The powder shall be dry, white to light buff in color, free-flowing and free from lumps. No rust shall be observed in the interior of the container. The contents of the container shall then be subjected to the performance test specified in Table II and 4.3.2.

4.4 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with PPP-P-16.

5. PREPARATION AND DELIVERY

5.1 Preservation and packaging. The preparation of the developer for delivery, in 1- and 5-gallon sizes or other size required in 3.4 shall be in accordance with PPP-P-16, except as specified herein.

* 5.2 Unit packaging. Unless otherwise specified by the procuring activity, unit packaging shall be in accordance with PPP-P-16, Level A, Type II, Class 1. Commerical puncture type cans are also applicable. The container shall be constructed of a material that is inert to the chemicals used in the developer.

* 5.3 Unit container marking. Unless otherwise specified by the procuring activity, each unit container shall be durably and legibly labeled by lithographic or other suitable means. If gummed labels are used, they shall be affixed with water resistant adhesive. The adhesive shall not smear or blur the markings.

* 5.3.1 The following information shall be on the label:

a. Item identification.
b. Directions for mixing and use.
c. Manufacturers name and code number.
d. Net weight.

* 5.3.2 At the discretion of the supplier, the following shall be either lithographed directly on the label or clearly printed on the lid.

a. Specification number.
b. Type number.
c. Stock number.
d. Contract number.
e. Date of manufacture.

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* 5.3.3 In accordance with International Agreement (Sec 6.3), the final mixed volumes shall be indicated in US gallons or ounces, Imperial gallons or ounces, and litres or cubic centimeters. The expiration date shall be stated as: "To be used before...." Special storage conditions, if required, should also be indicated.

* 5.3.4 Date of manufacture. The date of manufacture shall be the month and year of the material specified herein. Example: (Feb 66) would refer to February 1966.

5.4 Packing. Packing shall be level A, B, or C of PPP-P-16, as specified in the contract or order (see 6.2).

5.5 Marking. Marking shall be in accordance with PPP-P-16. In addition, the following precautionary marking shall be included on the exterior container:

PHOTOGRAPHIC CHEMICALS
STORE IN A COOL DRY PLACE
AWAY FROM RADIO ACTIVE MATERIALS

6. NOTES

6.1 Intended use. The developer covered by this specification is intended for development of photographic continuous tone black and white sensitized paper.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.
b. Nominal size (1- or 5-gallon) unit package required or size specified by procuring activity (see 3.4).
c. Quantity required.
d. Applicable level of packing required (see 5.1 and 5.4).

* 6.3 International standardization agreement. Certain provisions of this specification (5.3) are the subject of international standardization agreement (ASCC Air Std. 52/3, STANAG 3187). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

* 6.4 The margin of this specification are marked with an asterisk to indicate where changes, (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 70
Navy - AS
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Reviewer activities:
Air Force - 11
Navy - MC
Army - EL
DGSC - GS

User activities:
Air Force -
Navy - SH, YD
Army - AV, MI

Preparing activity:
Air Force - 70

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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